

Synopsis

Machine Learning Based Diabetes Risk Prediction System Using Clinical Data

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Problem Description

Diabetes is a common and serious health condition that affects a large number of people worldwide. If it is not detected at an early stage, it can lead to major complications such as heart disease, kidney problems, and even loss of vision. Early diagnosis plays an important role in reducing these risks, but in many cases, proper testing requires laboratory facilities and expert medical consultation, which may not always be easily accessible or affordable.

In this project, we aim to develop a machine learning-based system that can help in predicting whether a person is diabetic or not using basic clinical data. The dataset used consists of 300 patient records and includes important features such as glucose level, blood pressure, BMI, insulin level, and age. By analysing these parameters, the system can identify patterns associated with diabetes.

The process involves data preprocessing, feature selection, model training, and evaluation using different classification algorithms.

The final output of the system is a simple prediction: 0 for non-diabetic and 1 for diabetic.

This system can be useful as a supportive tool for healthcare professionals by providing quick initial predictions and assisting in early diagnosis.

Dataset Description

The dataset used in this project is based on the Pima Indians Diabetes Dataset, which contains medical diagnostic measurements of patients. For this project, a dataset of 300 patient records is used, consisting of 8 input features and 1 target variable (Outcome).

Each record represents a patient and includes clinical attributes such as glucose level, blood pressure, BMI, insulin level, and age. The target variable indicates:

- 0 → Non-Diabetic
- 1 → Diabetic

Out of 300 records:

- 170 patients (56.7%) are diabetic
- 130 patients (43.3%) are non-diabetic

Sample Dataset (First Few Records)

Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPed	Outcome
2	120	70	25	80	28.5	0.45	1
4	150	85	32	130	33.2	0.67	1
1	95	60	20	70	24.1	0.30	0
3	180	90	35	160	36.5	0.80	1
0	105	65	22	75	26.8	0.40	0
5	165	88	30	140	34.7	0.72	1

Dataset Analysis and Key Insights

Before training the models, the dataset was analyzed to understand data quality and patterns.

- **Missing Values:**
Some features like Glucose, Blood Pressure, Skin Thickness, Insulin, and BMI had about 5% missing values, which were handled using median imputation.
- **Feature Importance:**
Glucose was found to be the most important feature, followed by BMI and Age, as they strongly influence diabetes prediction.
- **Feature Distribution:**
Glucose values range from around 50 to 338 mg/dL with an average near 122. BMI values show that many patients fall in the overweight range.
- **Class Distribution:**
The dataset is slightly imbalanced, with 56.7% diabetic and 43.3% non-diabetic cases, but it is still manageable.
- **Key Observation:**
Higher glucose levels and BMI are strongly linked to diabetes, and the risk increases with age.

Steps of Implementation

The project follows a structured machine learning pipeline:

1. Data Collection

The dataset contains 300 patient records with features like glucose level, blood pressure, BMI, insulin, skin thickness, age, and number of pregnancies. It is based on the Pima Indians Diabetes Dataset and stored in a CSV file, which is loaded using Pandas in Jupyter Notebook. The dataset was initially explored by checking its shape and viewing sample rows to understand the columns.

2. Data Preprocessing

Preprocessing was done to improve data quality. Some features had around 5% missing values, which were filled using median imputation. Outliers were removed using the IQR method. Finally, normalization and standardization were applied to scale the data, which is important for models like KNN and SVM.

- Missing values handled using median
- Outliers removed using IQR
- Feature scaling applied

3. Feature Selection

Correlation analysis was used to identify important features. Glucose was found to be the most important, followed by BMI and Age. Features with low impact were removed to reduce complexity. A heatmap was used to visualize feature relationships.

Important features: Glucose, BMI, Age

4. Model Selection and Training

Three models were used:

- Logistic Regression: Simple baseline model
- KNN: Based on nearest neighbors
- SVM: Finds the best decision boundary

The dataset was split into 80% training and 20% testing. Models were trained on the training data and evaluated on the test data to avoid overfitting.

5. Model Evaluation

➤ Accuracy

It is used in machine learning that measures the proportion of correct predictions (both true positives and true negatives) out of the total number of predictions made. Accuracy formula = (Correct predictions) / (Total predictions).

➤ Precision

- It measures the accuracy of positive predictions, calculated as the ratio of true positives to all predicted positives
- Precision formula = $TP / (TP + FP)$. Where:

TP = True Positive. , TN = True Negative
FP = False Positive. , FN = False Negative

➤ Recall (Sensitivity)

It measures the ability of the model to correctly identify all positive instances.

Recall formula= $TP / (TP + FN)$.

➤ F1-Score

- It is used for calculating the harmonic mean of precision and recall.
- F1-score formula = $2 \times (Precision \times Recall) / (Precision + Recall)$.

➤ Confusion Matrix

- A confusion matrix is a table showing actual vs predicted values (TP, TN, FP, FN).
- It helps visualize model performance and types of errors.

	Predicted Positive	Predicted Negative
Actual Positive	TP (True Positive)	FN (False Negative)
Actual Negative	FP (False Positive)	TN (True Negative)

6. Prediction System

The final step is to build a simple prediction system where patient details like glucose, BMI, and age are given as input, and the model predicts whether the person is diabetic (1) or non-diabetic (0). The best-performing model is used for this. This system can help doctors or health workers get a quick initial screening before detailed medical tests.

Expected Modules / Libraries to be Used

Inbuilt Libraries:

- NumPy – Used for numerical operations like mean, median, and array handling.
- Pandas – Used for loading the dataset, data cleaning, and basic analysis.
- Matplotlib – Used for plotting basic graphs like histograms.
- Seaborn – Used for advanced visualizations like heatmaps and pair plots.
- Scikit-learn (sklearn) – Used for building models, splitting data, and evaluating performance.

Custom Modules:

- Data Preprocessing Module – Handles missing values, removes outliers, and applies scaling to prepare clean data.
- Feature Selection Module – Identifies important features using correlation analysis.
- Model Training Module – Trains models like Logistic Regression, KNN, and SVM.
- Model Evaluation Module – Evaluates models using accuracy, precision, recall, and F1-score.
- Prediction Module – Takes input data and predicts whether the patient is diabetic or not.

Expected Evaluation Metrics and Plots

Evaluation Metrics

- Accuracy – Overall correctness of predictions across both classes.
- Precision – Measures how reliable the positive (diabetic) predictions are.
- Recall (Sensitivity) – Measures how well the model catches actual diabetic cases – this matters a lot in healthcare.
- F1-Score – Useful because the dataset has a slight imbalance (56.7% diabetic vs 43.3% non-diabetic).
- Confusion Matrix – Breaks down predictions into TP, TN, FP, FN for a deeper look at errors.

Visualization Plots

- ROC Curve – To compare the three models and check which one performs better across different thresholds.
- Heatmap – To visualize feature correlations and understand dependencies in the data.
- Pair Plot – Helps in seeing how features like Glucose and BMI relate to the target class when plotted against each other.

Required Topics from the Subject

- Arrays and Data Structures – Used to store and process data using NumPy arrays and Pandas DataFrames.
- Functions – Used to organize code into reusable steps like preprocessing, training, evaluation, and prediction.
- File Handling – Dataset is loaded using `read_csv()`, and results are saved for analysis.
- Basic Statistics – Mean, median, standard deviation, and correlation are used to understand the data.

Platform Required

- Python
- Jupyter Notebook
- Visual Studio Code

Books and Link Sources

1. <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>
2. <https://scikit-learn.org>
3. <https://pandas.pydata.org>
4. <https://www.geeksforgeeks.org/machine-learning/>

Final Output

- 0 → Non-Diabetic
- 1 → Diabetic